

The Recursion Theorem

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Relational structures

Relational structures, intuitively

Often in mathematics (including metalogic), we are interested in bundles comprising a set D together with one or more relations on D (which could be regular binary relations on D , i.e. subsets of $D \times D$, or ternary relations (subsets of $D^3 = (D \times D) \times D \dots$, or $\dots$

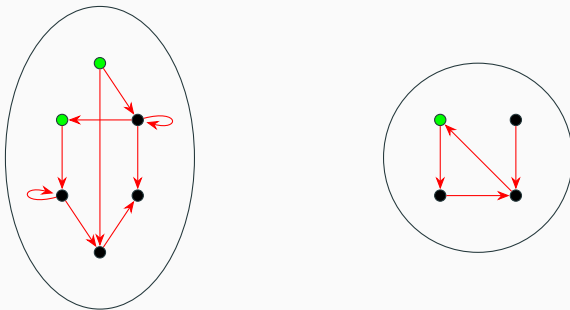
- When we talk of ' n -ary relations' on a set, we include the case of *singular* (1-ary) relations on a set as a special case: 'singular relation on A ' is just another way of saying 'subset of A '. We'll identify the 'range' of a subset with the subset itself.

The most common way to make sense of such "bundles" is just to look at a tuple where the first co-ordinate is the set A and the remaining co-ordinates are the relations on A that we are interested in.

Example: DXR-bundles

For example, we could consider triples of the form $\langle D, X, R \rangle$ where D is a set, X is a subset of D , and R is a binary relation on D . Let's call such triples “DXR-bundles” as a mnemonic. (We won't use this terminology after today).

We can picture a DXR-bundle as a set where some elements may be colored green (the ones in X) and where there may be arrows between some elements (the pairs in R).

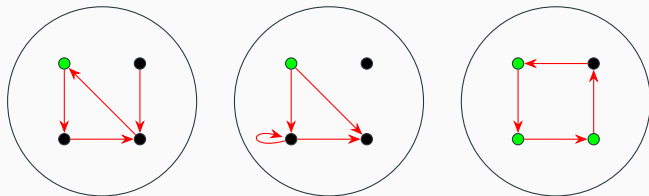


Some properties of DXR-bundles

Definition

A DXR-bundle $S = \langle D, X, R \rangle$ is *algebraic* iff (i) X is a singleton and (ii) R is a function.

The DXR-bundle pictured on the left is algebraic, while the ones on the right are not.



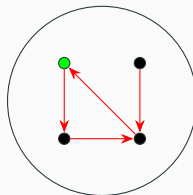
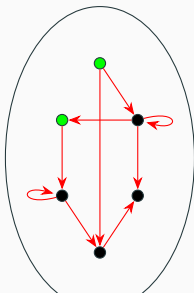
Some properties of DXR-bundles

Definition

A DXR-bundle $S = \langle D, X, R \rangle$ has the *Inductive Property* iff D is the closure of X under R .

That is: for every $C \subseteq D$, if $x \in C$ whenever $x \in X$, and $y \in C$ whenever $x \in C$ and Rxy , then $C = D$.

The DXR-bundle on the left has the Inductive Property; the one on the right doesn't



Definition

A DXR-bundle $S = \langle D, X, R \rangle$ has the *Injective Property* iff (i) R is injective, and (ii) no element of X is in the range of R .

Definition

Suppose $S = \langle D, X, R \rangle$ and $S' = \langle D', X', R' \rangle$ are DXR-bundles. Then a *homomorphism from S to S'* is any function $h : D \rightarrow D'$ such that

- (i) whenever $x \in X$, $hx \in X'$
- (ii) whenever Rxy , $R'(hx)(hy)$.

The Recursion Theorem for DXR-bundles

Using the special DXR-bundle terminology introduced on the last few slides, we can state the following theorem. (We won't prove it yet since it'll be more useful to prove a more general theorem of which it's a special case.)

Recursion Theorem for DXR-bundles

Suppose $S = \langle D, X, R \rangle$ is a DXR-bundle with the Inductive and Injective Properties and $S' = \langle D', X', R' \rangle$ is an algebraic DXR-bundles. Then there is a unique homomorphism from S to S' .

The Recursion Theorems for DXR-bundles and Numbers

We can consider the natural numbers as a DXR-bundle $\langle \mathbb{N}, \{0\}, \text{suc} \rangle$. It is algebraic, and has the Inductive and Injective properties. So as a special case of the Recursion Theorem for DXR-bundles, we have that there is a unique homomorphism from $\langle \mathbb{N}, \{0\}, \text{suc} \rangle$ to any algebraic DXR-bundle.

In other words:

Recursion Theorem for Numbers

Suppose D is a set, $z \in D$, and $s : D \rightarrow D$. Then there is a unique function $f : \mathbb{N} \rightarrow D$ such that $f0 = z$ and for all $n \in \mathbb{N}$, $f(\text{suc } n) = s(fn)$.

The supposition can be restated as ' $\langle D, \{z\}, s \rangle$ is an algebraic DXR-bundle'. And the property singling out f is equivalent to ' $fx \in \{z\}$ whenever $x \in \{0\}$, and whenever $\langle n, m \rangle \in \text{suc}$, $\langle fn, fm \rangle \in s$ ', in other words ' f is a homomorphism from $\langle \mathbb{N}, \{0\}, \text{suc} \rangle$ to $\langle D, \{z\}, s \rangle$ '.

Proving the Recursion Theorem for DXR-bundles

Let's prove the Recursion Theorem for DXR-bundles, since the idea of the proof is exactly the same as that of the more general Recursion Theorem we'll have later.

Proof: Let $X^+ = X \times X'$.

Let R^+ be the binary relation on $D \times D'$ such that $R^+\langle x, x' \rangle \langle y, y' \rangle$ iff Rxy and $R'x'y'$.

Let F be the closure of X^+ under R^+ .

We will first prove that F is a function from D to D' , and then verify that it meets the further conditions required to be a homomorphism from S to S' . Finally we'll show that there is at most one homomorphism from S to S' .

Proving the Recursion Theorem

(i) F is serial, i.e. for all $x \in D'$ there exists $x' \in D'$ such that Fxx' . By induction.

Base case: if $x \in X$, then since X' is nonempty, there is some $x' \in D'$ such that $\langle x, x' \rangle \in X^+ \subseteq F$.

Induction step: Suppose Fxx' and Rxy . Since R' is serial, there exists y' such that $R'x'y'$. But then $R^+\langle x, x' \rangle \langle y, y' \rangle$, so Fyy' since F is closed under R^+ .

Proving the Recursion Theorem

(ii) F is functional, i.e. for any $x \in D$, if Fxx' and Fxx'' then $x' = x''$. By induction.

Base case: suppose $x \in X$. Then since x isn't in the range of R , no ordered pair $\langle x, x' \rangle$ is in the range of any R^+ . So the only way $\langle x, x' \rangle$ and $\langle x, x'' \rangle$ could both be in F is if both are in $X^+ = X \times X'$. But since X' is a singleton, it follows that $x' = x''$.

Induction step: suppose that x is F -related to a unique x' , Rxy , and Fyy' and Fyy'' ; we will show that $y' = y''$. Since $y \notin X$, neither $\langle y, y' \rangle$ nor $\langle y, y'' \rangle$ is in X^+ , so both must be in the range of R^+ : i.e. we have $R^+\langle z, z' \rangle \langle y, y' \rangle$ and $R^+\langle u, u' \rangle \langle y, y'' \rangle$ for some $\langle z, z' \rangle$ and $\langle u, u' \rangle$ in F . But then Rzy and Ruy , so $z = u = x$ since R is injective. So by the induction hypothesis, $z' = u' = x'$. Thus we have $R'x'y'$ and $R'x'y''$, hence $y = y''$ by the fact that R' is functional.

Proving the Recursion Theorem

(iii) F is a homomorphism from S to S' .

Suppose $x \in X$ and Fxx' . Then since x is not in the range of R , $\langle x, x' \rangle$ is not in the range of R^+ , so it must be in X^+ , so $x \in X'$.

Suppose Rxy , Fxx' , and Fyy' . Then since y is not in X , $\langle y, y' \rangle$ is not in X^+ , so there must be some $\langle z, z' \rangle \in F$ such that $R^+\langle z, z' \rangle \langle y, y' \rangle$, i.e. Rzy and $R'z'y'$. But since R is injective we must have $z = x$, and thus since F is functional we must have $z' = x'$, so $R'x'y'$.

Proving the Recursion Theorem

(iv) There is at most one homomorphism from S to S' .

Suppose g and h are homomorphisms from S to S' : we prove by induction that $gx = hx$ for all $x \in D$.

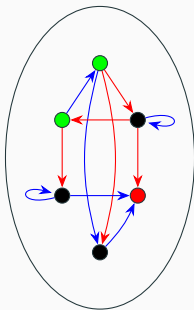
Base case: $x \in X$. Then gx and hx must both be in X' , so $gx = hx$ since X' is a singleton.

Induction step: suppose $gx = hx$ and Rxy . Then $R'(gx)(gy)$ and $R'(gx)(hy)$; but then $gy = hy$ since R' is functional.

Example: DXYRS-bundles

As another example, we could consider quintuples of the form $\langle D, X, Y, R, S \rangle$, where D is a set, X and Y are subsets of D , and R and S are binary relations on D : let's call these 'DXYRS-bundles'.

We can picture a DXYRS-bundle as a set where some elements may be colored green (X) or red (Y) or both, and where any pair of elements may be connected by a red (R) or blue (S) arrow or both:



Example: DXYRS-bundles

All the definitions that we had for DXR-bundles carry over to DXYRS-bundles. For a DXYRS-bundle $B = \langle D, X, Y, R, S \rangle$:

- ▶ B is algebraic iff X and Y are singletons and R and S are functions.
- ▶ B has the Inductive Property iff D is the closure of $X \cup Y$ under R and S .
- ▶ B has the Injective Property iff R and S are injective and no two of X , Y , $\text{range } R$, and $\text{range } S$ overlap.

For two DXYRS bundles $B = \langle D, X, Y, R, S \rangle$ and $B' = \langle D', X', Y', R', S' \rangle$:

- ▶ A homomorphism from B to B' is a function $h : D \rightarrow D'$ such that $hx \in X'$ whenever $x \in X$, $hx \in Y'$ whenever $x \in Y$, $R'(hx)(hy)$ whenever Rxy , and $S'(hx)(hy)$ whenever Sxy .

And we have a Recursion Theorem for DXYRS-bundles too: if B has the Inductive and Injective properties and B' is algebraic, there is a unique homomorphism from B to B' .

Generalizing

We need a flexible way of generalizing these concepts to “bundles” that build in any number of relations of any given “arity”. Here’s how we’ll do it.

Definition

A *relational signature* Σ is an ordered pair $\langle I, (a_i)_{i \in I} \rangle$, where I_Σ is a set and for each i , $a_i \in \mathbb{N} \setminus \{0\}$.

The idea is that each such Σ will correspond to a particular “type” of relational structure, which contains as many n -ary relations as there are elements of I for which $a_i = n$. The elements of I are just arbitrary labels.

Notation: Sometimes we’ll give a function a name like $(a_i)_{i \in I}$ or just (a_i) , where I is the function’s domain. When we pick a name like for a function, we get to write ‘ a_i ’ (with a subscript) to denote the value of the function on a particular argument $i \in I$. This makes things easier to read, and is convenient when we are just using the set I as an arbitrary collection of labels rather than caring about it for its own sake.

Relational structures, officially

Definition

Where $\Sigma = \langle I, (a_i) \rangle$ is a relational signature, a *relational structure* for Σ is a pair $S = \langle D, (R_i) \rangle$, where for each $i \in I$, $R_i \subseteq D^{a_i}$.

- ▶ The elements of I can be anything we like, but it might be helpful to think of them as sets of *colors*. Then we can diagram a relational structure for Σ by a collection of dots where each dot can have one or more of the colors i for which $a_i = 1$; each pair of dots can be connected by an arrow with any of the colors i for which $a_i = 2$; etc.
- ▶ It'll often be convenient to use sets of numbers: e.g. we can identify DXR-bundles with structures for the signature where $I = \{1, 2\}$, $a_1 = 1$, $a_2 = 2$, and DXYRS-bundles with structures for the signature where $I = \{1, 2, 3, 4\}$, $a_1 = a_2 = 1$, $a_3 = a_4 = 2$, etc.
- ▶ Later, we'll have special uses for signatures where I is a set of strings.

Definition

A relational structure $\langle D, (R_i) \rangle$ for Σ is an *algebraic structure* for iff R_i is a function whenever $a_i > 1$ and R_i is a singleton whenever $a_i = 1$.

- Why is it natural to group the two parts of the definition together? Well, an n -ary relation R on D is an $n - 1$ -ary function iff for all $x_1, \dots, x_{n-1} \in D$ there is exactly one x_n such that $Rx_1 \dots x_{n-1}x_n$.
- In the case of $n = 1$, 'for all $x_1, \dots, x_{n-1} \in D$ ' isn't adding anything; so we can identify *0-ary functions on D* with *singleton subsets of D* .

Definitions for relational structures

Definition

A relational structure $S = \langle D, (R_i) \rangle$ has the *inductive property* iff D is the only subset of D closed under each (R_i) .

- ▶ When R is an n -ary relation on D , $C \subseteq D$ is closed under R iff whenever $x_1, \dots, x_{n-1} \in C$ and $Rx_1 \dots x_{n-1}x_n$, $x_n \in C$.
- ▶ For $n = 1$, this just means $R \subseteq C$!
- ▶ So, another way to put this definition is: D is the closure of $\bigcup \{R_i \mid a_i = 1\}$ under all of the R_i for which $a_i > 1$.

Definitions for relational structures

Suppose we have a relational structure for Σ , $\langle D, (R_i) \rangle$, that has the Inductive Property, and we want to show that every element of D has a certain property ϕ . Then we can do so by a proof by induction.

In this kind of proof by induction, there is just one step: we consider R_i (where $a_i = n$) and suppose that $\phi(x_1), \dots, \phi(x_{n-1})$ and $Rx_1 \dots x_n$. Then we have to prove that $\phi(x_n)$. If we can do this for all i , we can conclude that every element of D is ϕ .

- Here we have a ‘base case’ for every singular R_i and an ‘induction step’ for every n -ary R_i where $n > 1$. But if we’re proving something about relational structures, for an arbitrary signature Σ , we won’t be listing these separately in our proof.

Definition

A relational structure $S = \langle D, (R_i) \rangle$ has the Injective Property iff

- (i) R_i is injective whenever $a_i > 1$.
- (ii) Whenever $i \neq j$, the ranges of R_i and R_j do not overlap.

- We identify the “range” of a set considered as a singular relation with the set itself.

Homomorphisms of relational structures

A homomorphism from one relational structure to another will be a function from the carrier set of the first to the carrier set of the second that preserves all the relations.

That is:

Definition

Suppose $S = \langle D, (R_i) \rangle$ and $S' = \langle D', (R'_i) \rangle$ are two relational structures for Σ . Then a *homomorphism* from S to S' is a function $h : D \rightarrow D'$ such that for each i , whenever $R_i x_1 \dots x_n$, $R'_i(hx_1) \dots (hx_n)$.

Homomorphisms of algebraic structures

When the relational structures we are interested in are both algebraic structures, the condition for a homomorphism can be restated as follows:

Fact

Suppose $S = \langle D, (f_i) \rangle$ and $S' = \langle D', (g_i) \rangle$ are two algebraic structures for Σ . Then h is a homomorphism from S to S' iff $h : D \rightarrow D'$ and for each i ,
$$h(f_i(x_1, \dots, x_{n-1})) = g_i(hx_1, \dots, hx_{n-1}).$$

Easy facts about homomorphisms

Fact

If h is a homomorphism from S to S' and g is a homomorphism from S' to S'' , $g \circ h$ is a homomorphism from S to S'' .

Fact

The identity function on the domain of a structure is a homomorphism from that structure to itself.

Definition

h is an *isomorphism* from S to S' iff h is a homomorphism from S to S' , h is a bijection, and h^{-1} is a homomorphism from S' to S .

Fact

When S and S' are algebraic structures, h is an isomorphism from S to S' iff h is a homomorphism from S to S' and h is a bijection.

Proof: Let $S = \langle D, (f_i) \rangle$ and $S' = \langle D', (g_i) \rangle$, and h a bijective homomorphism from S to S' . It suffices to show that h^{-1} is a homomorphism from S' to S . Fix i and $y_1, \dots, y_n \in D'$ (where $n = a_i$). Since h is a homomorphism we have

$$\begin{aligned} h(f_i(h^{-1}y_1, \dots, h^{-1}y_n)) &= g_i(h(h^{-1}y_1), \dots, h(h^{-1}y_n)) \\ &= g_i(y_1, \dots, y_n) \end{aligned}$$

and hence

$$f_i(h^{-1}y_1, \dots, h^{-1}y_n) = h^{-1}(g_i(y_1, \dots, y_n))$$

The Recursion Theorem for relational structures

The Recursion Theorem

The Recursion Theorem

Suppose that $S = \langle D, (R_i) \rangle$ and $S' = \langle D', (R'_i) \rangle$ are relational structures for a signature $\Sigma = \langle I, (a_i) \rangle$, such that S has the Injective and Inductive properties and S' is an algebraic structure.

Then there is a unique homomorphism from S to S' .

Proving the Recursion Theorem

Proof: For each i (where $a_i = n$), we define R_i^+ to be the following n -ary relation on $D \times D'$:

$$R_i^+ := \{ \langle \langle x_1, x'_1 \rangle, \dots, \langle x_n, x'_n \rangle \rangle \mid R x_1 \dots x_n \text{ and } R' x'_1 \dots x'_n \}.$$

Let F be the smallest subset of $D \times D'$ closed under each (R_i^+) .

We will first prove that F is a function from D to D' , then verify that it meets the further conditions required to be a homomorphism from S to S' , and finally prove that there is at most one homomorphism from S to S' .

Proving the Recursion Theorem

(i) F is serial, i.e. for all $x \in D'$ there exists $x' \in D'$ such that Fxx' . By induction (using S 's Inductive Property).

Suppose that $Fx_1x'_1 \dots$ and $Fx_{n-1}x'_{n-1}$ and $R_ix_1 \dots x_n$. Since R'_i is serial, there exists x'_n such that $R'_ix'_1 \dots x'_{n-1}x'_n$. But then $R_i^+ \langle x_1, x'_1 \rangle \dots \langle x_n, x'_n \rangle$, so Fyy' since F is closed under R_i^+ .

Proving the Recursion Theorem

(ii) F is functional, i.e. for any $x \in D$, if Fxx' and Fxx'' then $x' = x''$. By induction (using S 's Inductive Property). Suppose x_1 is F -related to a unique $x'_1, \dots$, and x_{n-1} is F -related to a unique x'_{n-1} ; $Rx_1 \dots x_n$; $Fx_n x'_n$; and $Fx_n x''_n$. We will show $x'_n = x''_n$.

By S 's Injective Property, x_n isn't in range R_j for $j \neq i$. So neither $\langle x_n, x'_n \rangle$ nor $\langle x_n, x''_n \rangle$ is in the range of any R^+j for $j \neq i$. So both are in the range of R_i^+ restricted to F —i.e. there exist $\langle y_1, y'_1 \rangle, \dots, \langle y_{n-1}, y'_{n-1} \rangle, \langle z_1, z'_1 \rangle, \dots, \langle z_{n-1}, z'_{n-1} \rangle \in F$ such that

$$R_i^+ \langle y_1, y'_1 \rangle \dots \langle y_{n-1}, y'_{n-1} \rangle \langle x_n, x'_n \rangle \quad \text{and} \\ R_i^+ \langle z_1, z'_1 \rangle \dots \langle z_{n-1}, z'_{n-1} \rangle \langle x_n, x''_n \rangle$$

But then we have $R_i y_1 \dots y_{n-1} x_n$ and $R_i z_1 \dots z_{n-1} x_n$. So by S 's Injective Property, each $y_k = z_k = x_k$. Then by the IH, each $y'_k = z'_k = x'_k$. So we have $R'_i x'_1 \dots x'_{n-1} x'_n$ and $R'_i x'_1 \dots x'_{n-1} x''_n$. Hence $x'_n = x''_n$ since R'_i is functional.

Proving the Recursion Theorem

(iii) F is a homomorphism. Suppose $Fx_1x'_1, \dots, Fx_nx'_n$, and $R_ix_1 \dots x_n$. Since x_n isn't in the range of R_j for any $j \neq i$, $\langle x_n, x'_n \rangle$ isn't in the range of R_j^+ for any $j \neq i$, so there must be $\langle y_1, y'_1 \rangle, \dots, \langle y_{n-1}, y'_{n-1} \rangle \in F$ such that

$$R_i^+ \langle y_1, y'_1 \rangle \dots \langle y_{n-1}, y'_{n-1} \rangle \langle x_n, x'_n \rangle$$

But then $R^i y_1 \dots y_{n-1} x_n$. So, each $y_k = x_k$ because R^i is injective, which implies each $y'_k = x'_k$ because F is functional.

Thus we have $R_i^+ \langle x_1, x'_1 \rangle \dots \langle x_n, x'_n \rangle$ and hence $R'_i x'_1 \dots x'_n$.

Proving the Recursion Theorem

(iv) There is at most one homomorphism from S to S' .

Suppose g and h are both homomorphisms from S to S' . Using S 's Inductive Property, we'll prove by induction that $gx = hx$ for every $x \in D$.

Suppose for induction that $gx_1 = hx_1, \dots$, and $gx_{n-1} = hx_{n-1}$, and $Rx_1 \dots x_{n-1}x_n$.

Then $R'(gx_1) \dots (gx_{n-1})(gx_n)$ and $R'(gx_1) \dots (gx_{n-1})(hx_n)$ since g and h are homomorphisms.

Since R' is functional, it follows that $gx_n = hx_n$.